

Module 06: Learning in Embodied

Learning Objectives

1. Define **Learning** within the context of Embodied.
2. Analyze how Learning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Learning.

Introduction

This module explores **Learning**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Learning is a critical component of the 8-part Active Inference spine, bridging the gap between Action and Communication.

Key Concepts

1. Learning as a Markov Blanket Boundary

How does Learning define the boundary between the agent and the environment?

2. Generative Models of Learning

What parameters involved in Learning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Learning drive the perception-action loop?

Applications

In Embodied, we see Learning manifest in: * **Specific Example 1:** A child learning to ride a bicycle initially over-corrects every wobble, yanking the handlebars back and forth in large,

jerky movements; over days of practice, the corrections become imperceptibly small and automatic as the generative model learns the precise mapping between vestibular tilt signals and the steering adjustments needed to maintain balance -- the transition from effortful wobbling to effortless gliding is the progressive reduction of sensorimotor prediction error through embodied learning. * **Specific Example 2:** A ballet dancer spending years at the barre progressively refines her arabesque until the position feels natural rather than strained; this learning trajectory shows the generative model absorbing the biomechanics of the pose into its motor priors, so that what once required conscious attention to hip rotation, spinal extension, and ankle placement becomes a single integrated prediction that the body enacts as a unified movement gestalt -- the hallmark of skilled motor learning as free energy minimization in the movement domain.

Conclusion

Understanding Learning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.